A direct split-prime bound for the tenth discriminant

Version 3, expanded exposition: $q < 10^6$

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Abstract

Assuming the determinant construction and Lemmas 4.2, 5.1, and 6.2 of Banks, we obtain $q < 10^6$ for imaginary quadratic fields of class number one. This improves the preceding cutoff $5 \cdot 10^7$ by a factor of 50. The key ingredient is a direct upper bound on the weighted sum of split primes, obtained by counting primitive lattice points. It replaces both the theta-integral zero construction and the zero-dependent prime-bias estimate. Banks's determinant argument remains an external input. This expanded version supplies the background, determinant notation, and finite verification within one paper; no earlier version is needed. The bound and analytic constants are unchanged. No Heegner method or small-conductor zero computation is used.

Status. This is an AI-assisted research draft. Exact arithmetic checks and finite computations are supplied, but an independent mathematical cross-check of the v3 argument is pending. The completed check of v2 does not cover this argument. The original v0.3.0 paper remains available unchanged.

1 The problem, the result, and the route through the proof

Tao's tenth-discriminant challenge [3] asks how large a further imaginary quadratic field of class number one could be, using arguments related to Siegel zeroes rather than Heegner's solution of the class-number-one problem. Our aim is first to obtain an explicit analytic cutoff, and then to check every smaller candidate by elementary integer arithmetic. The two stages are distinct: the analytic argument supplies the bound 10^6; the finite check leaves exactly the nine familiar fields.

Write $K = \mathbb{Q}(\sqrt{d})$, where $d < 0$ is squarefree. Its field discriminant is the *fundamental discriminant*

$$D = \begin{cases} d, & d \equiv 1 \pmod 4, \\ 4d, & d \equiv 2, 3 \pmod 4. \end{cases}$$

Put

$$q = |D|, \qquad K = \mathbb{Q}(\sqrt{D}), \qquad R = \sqrt{q}, \qquad L = \log q.$$

Thus q is the absolute field discriminant; for example, $\mathbb{Q}(i)$ has $q = 4$. All logarithms are natural. The class number $h(K)$ counts ideal classes. We use the equivalence

$$h(K) = 1 \quad \Longleftrightarrow \quad \text{every nonzero integral ideal of } \mathcal{O}_K \text{ is principal.}$$

Write $\chi_D(n) = (D/n)$ for the quadratic Kronecker character of conductor q. The meaning of its prime values is explained below.

Theorem 1. *Assume Banks's determinant construction and Lemmas 4.2, 5.1, and 6.2 in [1], including the interpolation results used in that construction. Then*

$$h(K) = 1 \quad \Longrightarrow \quad q < 1\,000\,000.$$

Together with the finite verification in Section 9, this gives

$$h(K) = 1 \quad \Longleftrightarrow \quad q \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Here is the route through the proof.

1. Class number one turns prime ideals of norm p into algebraic integers of norm p. A split prime supplies two such lattice points in the upper half-plane.
2. A geometric count, followed by a small inclusion–exclusion sieve, bounds the weighted sum of split primes. Subtracting split and ramified primes from an explicit total prime sum gives many inert primes.
3. Banks's construction supplies a nonzero integer J. Its absolute size has an upper bound, while each inert prime forces a large prime-power divisor of J. For $q \geq 10^6$, the lower bound exceeds the upper bound.
4. Below 10^6, a separate exhaustive residue check excludes all but nine discriminants. Reduced binary quadratic forms verify that each survivor actually has class number one.

The standard background is quadratic ideal theory and reduction of positive binary quadratic forms; the facts needed from both are stated in this paper. We also use the explicit prime estimates of Rosser–Schoenfeld, stated in Section 5. Section 6 defines the determinant and states the exact external bounds used from Banks. It does not reprove his interpolation theorem. The present proof constructs no zero of an L-function and does not use Banks's Theorem 1.1, Lemma 2.1, or Corollary 6.3.

2 Quadratic integers, prime ideals, and the norm lattice

Embed K in $\mathbb{C}$, choosing $\sqrt{D} = i\sqrt{q}$. The ring of integers has the explicit basis

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega, \qquad \omega = \begin{cases} (1 + \sqrt{D})/2, & D \text{ odd}, \\ \sqrt{D}/2, & D \text{ even}. \end{cases}$$

In particular, $\operatorname{Re}\omega = u \in \{0, 1/2\}$ and $\operatorname{Im}\omega = R/2$. The field norm is $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \alpha\bar{\alpha} = |\alpha|^2$. For integer coordinates a, b, write

$$Q(a, b) := \mathrm{N}_{K/\mathbb{Q}}(a + b\omega) = (a + ub)^2 + \frac{q}{4}b^2.$$

It is a positive integer for $(a, b) \neq (0, 0)$. An element is nonrational exactly when $b \neq 0$, so every nonrational integer has norm at least $q/4$. An ordinary integer a has norm a^2, which cannot be prime.

For an integral ideal $\mathfrak{a}$, its ideal norm is the index $\mathrm{N}\,\mathfrak{a} = [\mathcal{O}_K : \mathfrak{a}]$. For a principal ideal, this equals the field norm of a generator. The three possibilities for a rational prime p are

$\chi_D(p)$	factorization of $p\mathcal{O}_K$	norms of prime ideals
1	$\mathfrak{p}\,\bar{\mathfrak{p}}$, $\mathfrak{p} \neq \bar{\mathfrak{p}}$	p, p
-1	$p\mathcal{O}_K$ is prime	p^2
0	$\mathfrak{p}^2$	p

These are called split, inert, and ramified, respectively. Ramification means $p \mid D$. For odd p, $\chi_D(p)$ is zero if $p \mid D$, and otherwise is 1 or -1 according as D is a square or a nonsquare modulo p. For $p = 2$, it is zero when D is even, 1 when $D \equiv 1 \pmod 8$, and -1 when $D \equiv 5 \pmod 8$.

Lemma 2 (The small-prime obstruction)**.** *If $h(K) = 1$, every prime $p < q/4$ is inert.*

Proof. A split or ramified prime has an ideal of norm p. Under class number one, that ideal is (α) with $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = p$. The generator is nonrational, so the norm formula gives $p \geq q/4$, a contradiction. □

For $q > 4$, the only units are ± 1. Thus an ideal with one generator α has exactly the two generators $\alpha, -\alpha$. If its norm is prime, neither is real, and exactly one has positive imaginary part. Distinct split prime ideals therefore give exactly two points with $b > 0$ and norm p. This explains the factor $1/2$ in the split-prime estimate below.

3 Counting lattice points and imposing primitivity

For $x \geq 0$, define

$$A(x) = \#\{(a, b) \in \mathbb{Z} \times \mathbb{Z}_{\geq 1} : Q(a, b) \leq x\}.$$

The region $Q(a, b) \leq x$ is an ellipse in the coordinate plane. The restriction $b > 0$ counts one half of the nonrational points.

Lemma 3. *For every $x \geq 0$,*

$$\frac{\pi x}{R} - 2\sqrt{x} - \frac{2\sqrt{x}}{R} \leq A(x) \leq \frac{\pi x}{R} + \frac{2\sqrt{x}}{R}.$$

Proof. Put $B = 2\sqrt{x}/R$ and $f(t) = 2\sqrt{x - qt^2/4}$ on $[0, B]$. For each integer $1 \leq b \leq B$, the allowed real values of a form a closed interval of length $f(b)$. Its integer count lies between $f(b) - 1$ and $f(b) + 1$. Since f is decreasing,

$$\int_0^B f(t)\,dt - f(0) \leq \sum_{b=1}^{\lfloor B \rfloor} f(b) \leq \int_0^B f(t)\,dt.$$

The integral is $\pi x/R$, and there are at most B intervals. This gives both bounds, including when $B < 1$. □

A pair (a, b) is *primitive* if $\gcd(a, b) = 1$. A common divisor $d > 1$ would make d^2 divide $Q(a, b)$, so every pair of prime norm is primitive. We only remove common factors among four small primes. This leaves a superset of the primitive pairs, which is sufficient for an upper bound. Set

$$P = 2 \cdot 3 \cdot 5 \cdot 7 = 210, \qquad A_P(x) = \#\{(a, b) : b \geq 1,\ Q(a, b) \leq x,\ \gcd(a, b, P) = 1\}.$$

Lemma 4. *For every $x \geq 0$,*

$$A_P(x) \leq \frac{\pi\rho}{R} x + B_R\sqrt{x}, \qquad \rho = \frac{768}{1225}, \quad B_R = \frac{88}{35} + \frac{192}{35R}.$$

Proof. Write $\mu(d) = (-1)^k$ when $d \mid P$ has k prime factors. The pairs whose two coordinates are divisible by d correspond to the pairs counted by $A(x/d^2)$. Inclusion–exclusion therefore gives

$$A_P(x) = \sum_{d \mid P} \mu(d) A(x/d^2).$$

For positive coefficients use the upper bound of Lemma 3; for negative coefficients use its lower bound. The area coefficient is

$$\sum_{d \mid P} \frac{\mu(d)}{d^2} = \prod_{p \mid P} (1 - p^{-2}) = \rho.$$

Write

$$T = \sum_{d \mid P} \frac{|\mu(d)|}{d} = \frac{96}{35}, \qquad U = \sum_{d \mid P} \frac{\mu(d)}{d} = \frac{8}{35}.$$

The boundary coefficient is $2 \sum_{\mu(d)=-1} d^{-1} + 2T/R = T - U + 2T/R = B_R$. □

4 Direct bounds on split and ramified primes

Assume $h(K) = 1$ and $q \geq 10^6$, and put $a_0 = q/4$. If p splits, its two prime ideals of norm p are principal. As explained in Section 2, there are exactly four generators of norm p. They give two distinct pairs with $b > 0$. They are nonrational, and each satisfies $\gcd(a, b, P) = 1$, since $d \mid a, b$ would force $d^2 \mid Q(a, b) = p$. Consequently,

$$S_+(X) := \sum_{\substack{p \leq X \\ \chi_D(p) = 1}} \frac{\log p}{p} \leq \frac{1}{2} \sum_{\substack{b \geq 1,\ Q(a,b) \leq X \\ \gcd(a,b,P) = 1}} \frac{\log Q(a, b)}{Q(a, b)}. \tag{1}$$

All norms in this sum are at least a_0. For $f(t) = (\log t)/t$, we have $-f'(t) = (\log t - 1)/t^2 > 0$ on $[a_0, \infty)$. Partial summation expresses the weighted count as its endpoint term plus $-\int A_P(t) f'(t)\, dt$; the lower endpoint uses $A_P(a_0^-) = 0$, so points of norm exactly a_0 are included. For $X \geq a_0$, inserting Lemma 4 gives

$$\begin{aligned} S_+(X) &\leq \frac{1}{2} \left\{ A_P(X) \frac{\log X}{X} + \int_{a_0}^{X} A_P(t) \frac{\log t - 1}{t^2}\, dt \right\} \\ &\leq \frac{\pi \rho}{4R} \big((\log X)^2 - (\log a_0)^2 + 2 \log a_0 \big) + \frac{B_R (\log a_0 + 1)}{\sqrt{a_0}}. \end{aligned} \tag{2}$$

In the last line we dropped the nonpositive boundary term $-B_R(\log X + 2)/(2\sqrt{X})$. Here $\log a_0 > 2$, and $\log a_0 + 1 < L$ because $\log 4 > 1$. Also, for $R \geq 1000$,

$$\frac{\pi \rho}{4} < \frac{493}{1000}, \qquad B_R < \frac{63}{25}.$$

Thus the required direct split-prime bound is

$$S_+(X) < \frac{493}{1000R} (\log X)^2 + \frac{126L}{25R} \qquad (X \geq q/4). \tag{3}$$

A ramified prime likewise has a principal prime ideal of norm p. Its generator is nonrational, so $p \geq q/4$. Hence

$$S_0(X) := \sum_{\substack{p \leq X \\ p \mid q}} \frac{\log p}{p} \leq \frac{4}{q} \sum_{p \mid q} \log p \leq \frac{4L}{q}. \tag{4}$$

Neither this estimate nor the split-prime count requires a separate reduction to prime discriminants.

5 Obtaining the inert-prime mass

For any $X > H \geq 319$, the explicit prime estimates we use are

$$\sum_{H<p\leq X} \frac{\log p}{p} > \log X - \log H - \frac{1}{\log H}, \qquad \vartheta(X) := \sum_{p\leq X} \log p < 2X. \tag{5}$$

For completeness, Rosser–Schoenfeld's Theorem 6 [2] gives a constant C such that

$$\left| \sum_{p\leq x} \frac{\log p}{p} - \log x - C \right| < \frac{1}{2\log x} \qquad (x \geq 319).$$

Subtract the upper bound at H from the lower bound at X. Since $(2\log X)^{-1} < (2\log H)^{-1}$, this proves the first inequality in (5). Their Theorem 9 gives the stronger $\vartheta(x) < 1.01624x$ for $x > 0$, so also the second.

Every prime is split, inert, or ramified. Removing the full split and ramified sums up to X removes at least their contributions in $(H, X]$. Thus

$$\sum_{\substack{H<p\leq X \\ \chi_D(p)=-1}} \frac{\log p}{p} \geq \sum_{H<p\leq X} \frac{\log p}{p} - S_+(X) - S_0(X).$$

Combining (5), (3), and (4), for $X > H \geq 319$ and $X \geq q/4$ we obtain

$$\sum_{\substack{H<p\leq X \\ \chi_D(p)=-1}} \frac{\log p}{p} > \log X - \log H - \frac{1}{\log H} - \frac{493}{1000R}(\log X)^2 - \left(\frac{126}{25R} + \frac{4}{q}\right) L. \tag{6}$$

6 The determinant and the external results used

We now define the objects whose estimates are taken from Banks. This section is an explicit statement of that input, not a new proof of his interpolation result. Our notation K denotes the quadratic field; the biquadratic field called K in Banks is denoted by E here.

6.1 The points, the rows, and the nonzero integer

Let $d < 0$ be the squarefree integer from Section 1, and set

$$\eta = \sqrt{d}, \qquad \xi = \sqrt{2}, \qquad E = \mathbb{Q}(\eta, \xi), \qquad \mathcal{O} = \mathbb{Z}[\eta, \xi].$$

The field E has degree four over $\mathbb{Q}$. Its four automorphisms are $1, \sigma, \tau, \sigma\tau$, where σ changes the sign of η and τ changes the sign of ξ. Elements of $\mathcal{O}$ are algebraic integers, so their field norms are integers.

Choose integers $1 \leq H \leq N$, and put $M = N^{4/3}$. The real number M need not be an integer. The column set is $\mathcal{I} = \{0, \ldots, N-1\}^4$, of cardinality N^4. For each $\boldsymbol{n} = (n_1, n_2, n_3, n_4) \in \mathcal{I}$, set

$$\theta_{\boldsymbol{n}} = n_1 + n_2\eta + n_3\xi + n_4\eta\xi.$$

For $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{Z}^3_{\geq 0}$, form the row

$$\boldsymbol{v}_{\boldsymbol{\alpha}} = \left(\theta_{\boldsymbol{n}}^{\alpha_1} \sigma(\theta_{\boldsymbol{n}})^{\alpha_2} \sigma\tau(\theta_{\boldsymbol{n}})^{\alpha_3}\right)_{\boldsymbol{n}\in\mathcal{I}}, \qquad w(\boldsymbol{\alpha}) = \alpha_1 + H\alpha_2 + H\alpha_3.$$

Here $z^0 = 1$, including at $z = 0$. Order the indices by increasing weight, breaking ties by any fixed rule, and retain a row if it is linearly independent over E of the earlier retained rows. Let $\mathcal{A}$ be the retained index set.

The interpolation and construction results in Banks's Sections 3–4 give $|\mathcal{A}| = N^4$ and a basis of the N^4-dimensional row space. Consequently, the following determinant and integer are nonzero:

$$\Delta = \det(\boldsymbol{v}_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathcal{A}} \in \mathcal{O} \setminus \{0\}, \qquad J = \mathrm{N}_{E/\mathbb{Q}}(\Delta) = \Delta\,\sigma(\Delta)\,\tau(\Delta)\,\sigma\tau(\Delta) \in \mathbb{Z} \setminus \{0\}.$$

Define the degree sums

$$S_1 = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} \alpha_1, \qquad S_{23} = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} (\alpha_2 + \alpha_3).$$

These are the quantities denoted by S_1, S_{23} below. The argument uses their bounds; it never numerically constructs this enormous determinant.

6.2 The three estimates

For $q \geq 9$ and $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$, the precise consequences of the cited lemmas are as follows.

1. *Degree estimates (Lemma 4.2):*

$$S_1 \geq \frac{N^4 H^{2/3} M}{4 \cdot 97^2} > 0, \qquad S_{23} \leq 192 N^4 H^{-1/3} M. \tag{7}$$

2. *Absolute-size estimate (Lemma 5.1):*

$$\frac{1}{4}\log|J| \leq (S_1 + S_{23})\left(\log N + \frac{L}{2} + \log 8\right) + 2N^4 \log N. \tag{8}$$

   This bounds the determinant at its four complex embeddings.

3. *Prime-power divisibility (Lemma 6.2):* for every odd prime $p > H$ with $\chi_D(p) = -1$,

$$p^{4E_p} \mid J, \qquad E_p = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} \left\lfloor \frac{\alpha_1}{p} \right\rfloor. \tag{9}$$

   The factor four comes from taking the norm from E: a factor p^{E_p} of Δ contributes p^{4E_p} to its norm.

The mechanism in the last estimate is the Frobenius congruence $\theta^p \equiv g_p(\theta) \pmod{p\mathcal{O}}$, where g_p is σ or $\sigma\tau$. In a row, exchanging p powers of the first variable for one power of either other variable decreases the weight when $p > H$. Banks uses the resulting row operations to extract the stated prime powers. No hypothesis about a real zero is present in (7)–(9).

7 Applying the determinant estimates

Take

$$H = 10^9, \qquad N = q^{13}, \qquad M = N^{4/3}, \qquad z = \frac{52}{3}, \qquad \log M = zL.$$

For $q \geq 10^6$, Banks's hypotheses $q \geq 9$, $1 \leq H \leq N$, and $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$ hold, as do $M > H$ and $M \geq q/4$. All definitions and estimates of Section 6 therefore apply. Dividing the degree bounds (7) gives

$$\frac{S_{23}}{S_1} \leq \frac{768 \cdot 97^2}{10^9} < r := \frac{73}{10000}, \qquad \frac{N^4 M}{S_1} \leq \frac{4 \cdot 97^2}{10^6} < c := \frac{19}{500}.$$

Divide (8) by S_1. Since $N^4/S_1 < c/M$, it gives

$$\frac{\log|J|}{4S_1} \le (1+r)\left(\log N + \frac{L}{2} + \log 8\right) + \frac{2c\log N}{M}. \tag{10}$$

There are N^4 retained indices. The elementary bound $\lfloor t \rfloor \ge t-1$ therefore gives

$$E_p \ge \frac{S_1}{p} - N^4.$$

The powers in (9) for different primes are coprime, so their product divides the nonzero integer J. Hence

$$\begin{aligned}\frac{\log|J|}{4S_1} &\ge \sum_{\substack{H<p\le M\\ \chi_D(p)=-1}} \frac{E_p}{S_1}\log p \\ &\ge \sum_{\substack{H<p\le M\\ \chi_D(p)=-1}} \frac{\log p}{p} - \frac{N^4}{S_1}\sum_{\substack{H<p\le M\\ \chi_D(p)=-1}} \log p.\end{aligned}$$

The last sum is at most $\vartheta(M) < 2M$, and $N^4M/S_1 < c$. Consequently,

$$\frac{\log|J|}{4S_1} \ge \sum_{\substack{H<p\le M\\ \chi_D(p)=-1}} \frac{\log p}{p} - 2c. \tag{11}$$

This derives the lower bound from prime-power divisibility itself.

Compare (10) with (11) and insert (6) at $X = M$. The leading lower term is $\log M$, whereas the leading upper term is $(1+r)\log N = \frac{3}{4}(1+r)\log M$. Moving the other terms to the right, using $\log N = 13L$, and dividing by L gives the necessary inequality

$$\begin{aligned}\left(\frac{1}{4} - \frac{3r}{4}\right) z \le{}& \frac{493}{1000}z^2\frac{L}{R} + \frac{1+r}{2} + \frac{126}{25R} + \frac{4}{q} \\ &+ \frac{\log H + (1+r)\log 8 + 1/\log H + 2c}{L} + \frac{26c}{M}.\end{aligned} \tag{12}$$

8 A uniform rational contradiction

The derivative of $(\log q)/\sqrt{q}$ is $q^{-3/2}(1 - \frac{1}{2}\log q)$, so the function decreases for $q > e^2$. Using $q \ge 10^6$, we have the uniform estimates

$$\frac{L}{R} < \frac{13.816}{1000}, \qquad L > 13.815, \qquad R \ge 1000, \qquad \frac{\log H}{L} \le \frac{3}{2},$$
$$\log 8 < 2.08, \qquad \frac{1}{\log H} < 0.049, \qquad \frac{26c}{M} < 10^{-6}.$$

The last inequality follows already from $M \ge 10^6$ and $26c = 0.988 < 1$. The logarithmic estimates are certified by rational Taylor bounds for the exponential function in the accompanying program.

The left side of (12) equals

$$\left(\frac{1}{4} - \frac{3}{4}\frac{73}{10000}\right)\frac{52}{3} = \frac{127153}{30000} > 4.238.$$

Its right side is strictly less than the rational number

$$\mathcal{R} = \frac{493}{1000}\left(\frac{52}{3}\right)^2 \frac{13.816}{1000} + \frac{1.0073}{2} + \frac{3}{2} + \frac{1.0073(2.08) + 0.049 + 0.076}{13.815} + \frac{5.04}{1000} + \frac{4}{10^6} + \frac{1}{10^6} < 4.216.$$

Every terminating decimal here denotes its exact rational value. More precisely, the certificate checks

$$\frac{127153}{30000} - \mathcal{R} = \frac{867907363}{38375000000} > 0.022.$$

Every term has been bounded uniformly for all $q \geq 10^6$, rather than only at the endpoint. This contradicts (12) and proves the analytic cutoff in Theorem 1.

9 The complete finite verification below the cutoff

We now justify the finite classification within this paper. It uses Lemma 2, not the determinant estimates. The finite discriminant search here is also distinct from the sieve by $2, 3, 5, 7$ in Section 3, which counts lattice points for one field.

9.1 Why these are all the candidates

If $q > 8$ and $h(K) = 1$, Lemma 2 applies to $p = 2$. The splitting rule in Section 2 then gives

$$D \equiv 5 \pmod 8, \qquad q \equiv 3 \pmod 8.$$

The fundamental negative discriminants with $q \leq 8$ are exactly $-3, -4, -7, -8$, by the definition in Section 1. Thus, after the analytic cutoff, every additional candidate lies in the finite set

$$11 \leq q < 1\,000\,000, \qquad q \equiv 3 \pmod 8.$$

For each odd prime p with $4p < q$, the integer $-q$ must be a nonzero nonsquare modulo p. We exclude q if it fails even one of these necessary conditions.

It is sufficient to use only primes $p \leq 109$. The direct scan checks all 124 999 integers in the displayed progression. It deliberately includes nonfundamental integers: omitting the squarefreeness test enlarges the candidate set and cannot discard a genuine field. Let $T_p = \{a^2 \bmod p : 0 \leq a < p\}$, including zero. The complete exclusion rule can be written as the following Python pseudocode:

```
primes = [odd primes at most 109, generated by trial division]
squares = {p: {a*a % p for a in range(p)} for p in primes}
survivors = []
for q in range(11, 1_000_000, 8):
    if all((-q) % p not in squares[p]
           for p in primes if 4*p < q):
        survivors.append(q)
```

The supplied executable program `verify_finite_range.py` implements this scan using only integer arithmetic. It gives exactly

$$\texttt{survivors} = [11, 19, 43, 67, 163].$$

Its output records how many candidates each first rejecting prime eliminates; those counts sum to 124 994. Adding the five survivors accounts for every scanned integer.

9.2 Why each surviving field has class number one

Passing the residue tests is only a necessary condition. To verify the survivors, use the classical correspondence between ideal classes of $\mathcal{O}_K$ and proper equivalence classes of primitive positive definite integral binary quadratic forms

$$f(x,y) = ax^2 + bxy + cy^2, \qquad b^2 - 4ac = D = -q.$$

Here primitive means $\gcd(a,b,c) = 1$, and proper equivalence means an integral change of variables of determinant one. For example, the form determines the ideal class represented by $a\mathbb{Z} + \frac{-b+\sqrt{D}}{2}\mathbb{Z}$. The reduction theorem says that each class has exactly one representative with

$$|b| \le a \le c, \qquad b \ge 0 \text{ if } |b| = a \text{ or } a = c.$$

Thus the number of these reduced forms is $h(K)$. The inequalities imply

$$q = 4ac - b^2 \ge 3a^2, \qquad 1 \le a \le \left\lfloor \sqrt{q/3} \right\rfloor.$$

This makes the check exhaustive: enumerate these a, then $-a \le b \le a$, and retain $c = (b^2+q)/(4a)$ exactly when it is an integer with $c \ge a$, the gcd condition holds, and the boundary convention holds.

The program checks that every survivor is a fundamental discriminant and enumerates its reduced forms. The complete result is

q	(a,b,c)	q	(a,b,c)	q	(a,b,c)
3	$(1,1,1)$	8	$(1,0,2)$	43	$(1,1,11)$
4	$(1,0,1)$	11	$(1,1,3)$	67	$(1,1,17)$
7	$(1,1,2)$	19	$(1,1,5)$	163	$(1,1,41)$

Each field has exactly one reduced form. This proves the remaining assertion of Theorem 1, subject to the explicitly stated analytic inputs and the reproducible finite computation.

9.3 Relation to the larger existing certificate

The bundle also retains `verify_sieve.cpp` and `verify_independent.py`, with their original outputs. They check the larger interval $q < 10^{10}$, which contains the present one. For $q \ge 437$, they first combine the conditions for $2,3,5,7,11,13,17$ into 1440 residue classes modulo $8\cdot3\cdot5\cdot7\cdot11\cdot13\cdot17 = 2042040$. They then test primes from 19 through 109. All are eligible there because $4 \cdot 109 < 437$. Of 7 051 769 candidates in that range, none survive. For $11 \le q < 437$, they apply only the eligible conditions $4p < q$, leaving the same five values. The C++ program builds residue classes by the Chinese remainder theorem; the Python program scans the initial classes and uses periodic bitsets. This is supporting evidence. The direct scan just described suffices to reproduce the finite part of the present proof.

10 Reproduction, verification scope, and version history

Run these commands from this version's directory:

```
python3 verify_direct_prime.py --sanity
python3 verify_finite_range.py
```

Both use only the Python standard library. The first records its output in `checks.txt` and passes 26 exact arithmetic checks and 22 finite sanity cases. The second checks the entire remaining candidate progression and the nine reduced-form counts; its output is in `finite_range_checks.txt`.

10.1 What the arithmetic certificate establishes

The 26 checks verify the sieve constants, determinant parameters, logarithmic estimates, and final rational contradiction. For $x \geq 0$ and an integer n with $n + 2 > x$, put $T_n(x) = \sum_{k=0}^{n} x^k/k!$. The bound used in the program is

$$T_n(x) \leq e^x \leq T_n(x) + \frac{x^{n+1}}{(n+1)!} \frac{1}{1 - x/(n+2)}.$$

Indeed, after the first omitted term, each ratio of successive terms is at most $x/(n+2) < 1$, so a geometric series bounds the tail. All quantities are exact fractions. With $n = 160$, the program certifies, for example, $e^{13.815} < 10^6 < e^{13.816}$, which proves the two logarithmic bounds. The classical inequality $\pi < 22/7$ is used to check the coefficient 0.493.

The 22 sanity cases comprise 15 finite lattice comparisons and seven prime-norm multiplicity checks. The lattice examples compare direct enumeration with row counting, inclusion–exclusion, and the geometric bounds. The prime checks cover all primes at most 2000 for each of $q = 7, 8, 11, 19, 43, 67, 163$. These examples test the formulas and implementation; they do not prove the general counting lemmas.

10.2 What changed, and what remains external

The original v3 cutoff $q < 10^6$ [4], all its analytic constants, and its 26-check arithmetic certificate are unchanged in v0.3.1. This revision adds the setup and roadmap, defines the determinant objects and states their input bounds, expands the intermediate deductions, and includes the finite classification and a direct verifier for the exact remaining range. The preceding v2 cutoff was $q < 5 \cdot 10^7$; its theta-integral route is historical context and is not needed here.

Banks's interpolation and determinant results, the Rosser–Schoenfeld prime estimates, and the stated classical quadratic theory remain mathematical inputs. The programs do not audit those results or formalize the general analytic deductions. No independent mathematical cross-check of v3 or Lean formalization is recorded here. Codex generated the v3 derivation, this expanded exposition, and the verification programs. The completed independent check of v2 remains scoped to v2. Source URLs and snapshot hashes are in `sources.json`; detailed review status is in `REVIEW.md`.

References

[1] W. D. Banks, *On the OpenAI proof that Landau–Siegel zeros do not exist*, arXiv:2610.15483v1, especially the determinant construction in Sections 3–4 and Lemmas 4.2, 5.1, and 6.2.

[2] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94, Theorems 6 and 9. doi:10.1215/ijm/1255631807.

[3] T. Tao, *An experimental AI-math challenge: using tenth-discriminant bounds to explore arguments ruling out Siegel zeroes*, challenge and discussion, 20 October 2026.

[4] *A direct split-prime bound for the tenth discriminant*, original v3 release v0.3.0, 20 October 2026.